

CogniHire: Auditing Résumé Claims Instead of Scoring Candidates

Integrating Evidence Grounding, Machine Learning, and Deep Learning for Accountable Technical Screening

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Automated hiring systems overwhelmingly emit a *score*: a scalar ranking a candidate against others. Such scores are difficult to explain, impossible for a candidate to contest, and an increasing regulatory liability under New York City Local Law 144, the Illinois Human Rights Act as amended by HB 3773, and the European Union Artificial Intelligence Act. This article presents CogniHire, a technical screening system built on the opposite premise — it *audits the claims a candidate makes about themselves* and never scores the candidate — and uses it as a vehicle for examining how three technology families combine in accountable screening: evidence grounding, machine learning (ML), and deep learning (DL).

We survey the architecture, failure landscape, and regulatory trajectory of automated screening, then examine each technology family in turn. Evidence grounding constrains a language model to *select* claim text and never to *author* it; measured across roughly 100,000 adversarial trials over 5,200 résumés, the gate rejected 100% of meaning-preserving paraphrases and 100% of negation traps while admitting 100% of verbatim claims. For ML we report a nine-model, six-family comparison under grouped cross-validation with significance testing, and a within-query diagnostic that separates matching ability from candidate-level quality — a distinction pooled metrics cannot make, and one that overturned our own initial hypothesis. For DL we show that a plausible authored similarity constant of 0.50 would have falsely rejected 41.4% of genuine identity matches, against 3.4% at a threshold calibrated on Labeled Faces in the Wild.

Synthesising these, we argue that the useful output of automated screening is not a judgement of the person but an audit of their claims, and that the recurring engineering move is converting policy into structure: an import boundary that fails the build, a type with no field to hold a fabricated value, a vocabulary ban that forbids a composite score. We report a weak result and a rejected component alongside strong ones, because a system that publishes only its successes offers no evidence that it evaluates itself honestly.

CCS Concepts: • **Social and professional topics** → **Employment issues**; • **Computing methodologies** → **Machine learning**; • **Information systems** → **Data mining**.

Additional Key Words and Phrases: Automated hiring, evidence grounding, claim verification, algorithmic accountability, model comparison, benchmark critique, face verification, threshold calibration, explainable AI

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1 Introduction

A technical interview ends and a form is filled in: “*Strong communication. Good Flutter knowledge. 8.7/10.*” Six months later the number is indefensible. Nobody can reconstruct which answer moved it from 8.4 to 8.7. The candidate cannot appeal it, because there is nothing specific to appeal. The recruiter cannot defend it, because no record exists of what

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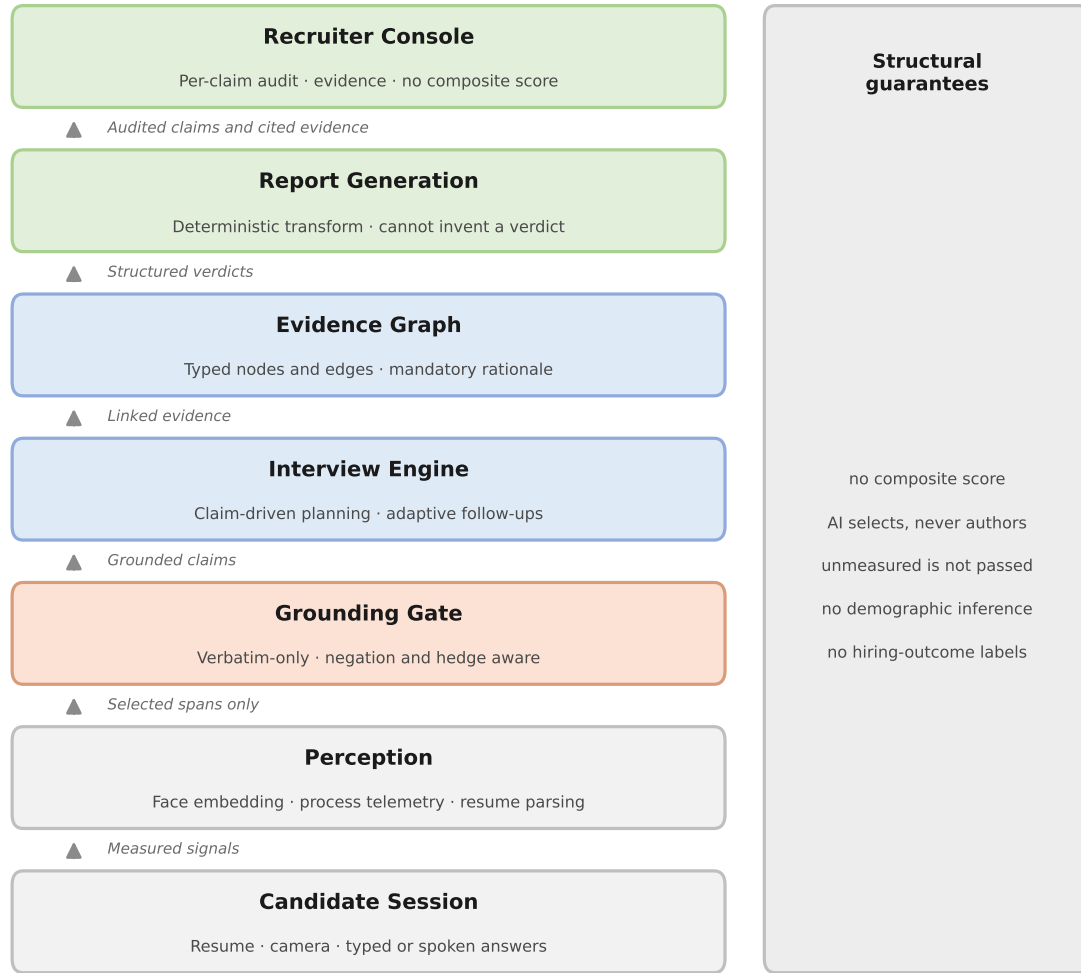


Fig. 1. Data flow and component interactions in CogniHire. Evidence moves upward from the candidate session to the recruiter console; each transition narrows what the system is permitted to assert. The right-hand band lists the guarantees enforced structurally rather than by convention (§7).

produced it. The number was never knowledge; it was a compression of knowledge that discarded everything needed to audit it.

Contemporary automated interview platforms largely reproduce this failure at scale. They add throughput and consistency and they retain the scalar. The industry’s response to criticism has been to make the scalar more sophisticated — more features, better models, fairness post-processing — rather than to ask whether a scalar was the right output object at all.

This article takes the opposite position. We argue that the useful output of an automated screening stage is not a judgement of the candidate but an **audit of the claims the candidate has made**. A résumé is a set of assertions. Each

assertion can be probed, each probe yields evidence, and evidence supports a verdict about *the claim*. The verdicts, with evidence attached, go to a human who decides about *the person*. Figure 1 shows the resulting data flow.

That reframing is not merely ethical positioning. It changes what the system must be able to do and, more importantly, what it must be *unable* to do — and it changes which technologies are appropriate at which layer. Three families carry the load. **Evidence grounding** constrains generative extraction so that a model may select a candidate’s own words but never compose new ones. **Machine learning** measures properties of the accumulated evidence. **Deep learning** supplies perception: face embedding for identity continuity, sentence encoding for semantic comparison. Each is examined in turn, and §7 examines what their combination buys that none delivers alone.

Scope of this article. We survey the architecture, failure landscape, and regulatory trajectory of automated candidate screening, and examine the integration of grounding, ML, and DL within it. Where the literature supports a claim we cite it; where we have measured something ourselves we report the measurement, the protocol, and the uncertainty. The system described is deployed, and §8 reports what deployment taught us, including three gaps we state rather than conceal.

1.1 Key Research Questions

This article addresses the following questions:

- (1) What distinct failure modes arise in automated candidate screening, and how do they differ from the failure modes of conventional information systems?
- (2) How does constraining a generative model to *selection* rather than *authorship* change the guarantees a screening system can offer, and at what measurable cost?
- (3) To what extent does the choice of machine-learning algorithm determine screening performance, once evaluation controls for leakage and multiple comparisons?
- (4) What does a pooled evaluation metric actually reward on a paired matching task, and can the components be separated?
- (5) How much does an authored constant differ from a calibrated one in a security-adjacent perception component, and how would anyone know?
- (6) What synergistic benefits emerge from integrating grounding, ML, and DL that none of the three provides in isolation?
- (7) How do regulatory instruments governing automated employment decisions map onto concrete architectural obligations?
- (8) Which open problems must be resolved before evidence-based screening can be deployed at scale, and which are research rather than engineering problems?

1.2 Structure of the Paper

Figure 2 outlines the organisation. Section 2, *Related Works*, reviews prior systems and surveys and positions this study’s contributions. Section 3, *Automated Hiring: Architecture, Failure Landscape, and Regulatory Trends*, discusses layered models, the failure taxonomy, and the regulatory response. Section 4, *Evidence Grounding as an Enabler*, presents the mechanism that makes the rest defensible. Sections 5 and 6, *ML and DL Techniques*, examine the learned components. Section 7, *Synergistic Integration*, assesses the combined stack. Section 8, *Deployment Experiences and Lessons*, presents what shipping it taught us, and Section 9, *Future Directions*, identifies open challenges. Section 10 concludes.

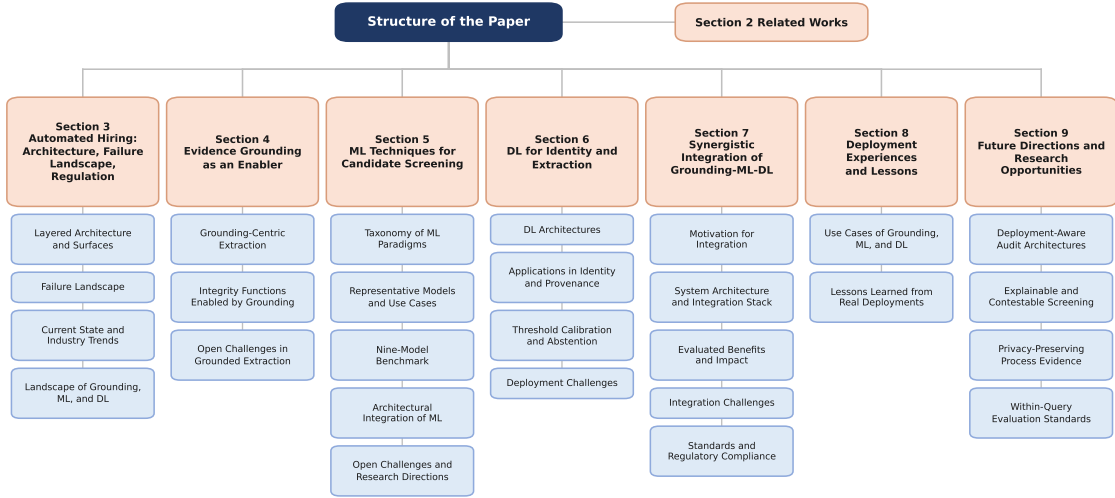


Fig. 2. Overview of the paper structure.

2 Related Works

2.1 Comparison of Existing Works

Work adjacent to accountable candidate screening divides into four groups: audits of deployed hiring systems, methodological critiques of model comparison, studies of leakage and evaluation protocol, and technical work on the perception components screening systems depend on. Most treat these concerns in isolation. Table 1 classifies the reviewed literature and states what this work adds.

Several strands warrant expansion. Raghavan et al. [14] surveyed vendors of algorithmic pre-employment assessment and drew attention to the consequences of a vendor’s choice of *prediction target*, since the fairness analysis that follows is conducted against the stated target rather than the deployed one. That concern is central here, and §5 supplies a measurement procedure for detecting when the two diverge.

A second strand re-ran published comparisons under controlled protocols and found the reported progress did not survive. Ferrari Dacrema et al. [4] examined eighteen neural recommendation methods, could reproduce only seven, and found six of those seven were often outperformed by simple heuristics. Musgrave et al. [10] reached an analogous conclusion for deep metric learning. Our §5 sits in this tradition but differs in target: rather than showing a complex method failing to beat a simple one, we find that *no* method separates from any other, and then trace why.

A third strand concerns evaluation protocol. Demšar [2] is the standard reference for comparing multiple classifiers and cautions specifically against reading a ranking off raw score differences. Gorman and Bedrick [5] showed system rankings established on a single standard split frequently fail to reproduce under regenerated splits. Kapoor and Narayanan [9] surveyed machine-learning-based science across seventeen fields and documented leakage affecting 294 papers. We adopt Demšar’s protocol and quantify one of Kapoor and Narayanan’s leakage categories on a hiring benchmark.

Some crucial *research gaps* follow from this comparison:

Table 1. Comparison of existing work on accountable screening and the research gaps addressed here. **G**: evidence grounding; **ML**: machine learning; **DL**: deep learning. ✓ = present, ✗ = not present.

Ref (Year)	Type	G	ML	DL	Focus	Gaps	Improvements in this work
[14] (2020)	*	✗	✓	✗	Vendor bias-mitigation claims	No measurement of what the target actually is	Within-query test detects target substitution
[4] (2019)	‡	✗	✓	✓	Neural recommender base-lines	Different domain; no grounding	Same protocol applied to screening
[10] (2020)	‡	✗	✗	✓	Metric-learning evaluation flaws	No deployed system	Protocol plus a shipped system
[2] (2006)	†	✗	✓	✗	Classifier comparison statistics	Method only	Applied end to end with power analysis
[5] (2019)	‡	✗	✓	✗	Standard-split fragility	NLP tagging domain	Grouped splits on a hiring corpus
[9] (2023)	*	✗	✓	✓	Leakage taxonomy across fields	No hiring instance	Leakage quantified on a hiring benchmark
[6] (2022)	‡	✗	✓	✓	Trees vs. deep nets on tabular data	Benchmarks only	Spread shown to be noise-dominated here
[3] (2019)	†	✗	✗	✓	Angular-margin face embedding	Threshold left to the integrator	Threshold calibrated and reported
[7] (2022)	†	✗	✗	✓	Efficient face detection	Detection only	Placed in a provenance pipeline
[8] (2007)	⊕	✗	✗	✓	Unconstrained verification benchmark	Not a hiring population	Used, with the mismatch stated
[13] (1999)	†	✗	✓	✗	Probability calibration	Classifier outputs	Applied to a similarity threshold
[15] (2005)	†	✗	✓	✗	Distribution-free abstention	Theory	Abstention as a hard gate
This work (2026)	—	✓	✓	✓	Grounded claim audit	—	Unified grounding + ML + DL, measured

⊕: benchmark/dataset; *: survey/review; ‡: applied research; †: framework/method.

- *Grounding is absent.* No reviewed work constrains generative extraction so that a model may select but not author. Extraction is treated as a generation problem and evaluated on fluency or coverage rather than on fidelity to the source document.
- *Prediction targets go unverified.* Vendors and benchmarks alike state a target and evaluate against a pooled metric that cannot distinguish it from an easier correlate.
- *Authored constants persist.* Perception components ship with similarity thresholds chosen by reasoning rather than measurement, and the error is invisible without a calibration study.
- *Isolation.* Grounding, ML, and DL are evaluated separately; no reviewed work reports what their integration measurably buys.

2.2 Our Contributions

- (1) A **claim-audit output model** replacing the composite score, with a four-valued verdict set that makes unmeasured evidence explicit rather than absent (§4).
- (2) A **structurally enforced grounding constraint** — the model may select claim text but never author it — implemented as an import boundary that a continuous-integration test enforces, and *measured* at 100% paraphrase and negation-trap rejection across roughly 100,000 trials (§4).

- (3) A **nine-model, six-family comparison** under grouped cross-validation with Friedman and Holm-corrected Wilcoxon testing, together with a synthetic positive control that establishes the protocol’s statistical power (§5).
- (4) A **within-query ranking diagnostic** separating matching ability from entity-level quality on any paired benchmark with a recurring entity (§5).
- (5) An **empirical threshold calibration** showing that a plausible, documented, authored constant was wrong in a specific and quantified way (§6).
- (6) A **measurement type system** in which a failed measurement carries no value, so a failure cannot be silently coerced into a pass (§7).
- (7) A **deployed implementation** with the gaps between design and deployment named rather than elided (§8).

2.3 Literature and Corpus Review Process

Two review processes underpin this article. For literature, every citation was retrieved and read against a primary source before inclusion; the project has previously had fabricated references caught in internal review, and the verification rule was adopted in response. Consequently the bibliography is smaller than is conventional for a survey of this length, and we prefer that to padding it with work we have not read.

For datasets, Figure 3 reports a screen of publicly hosted résumé-posting corpora against three requirements: independence from the corpus under study, recurring entities, and a binarisable fit label. The outcome is reported in §5 and is itself a finding.

3 Automated Hiring: Architecture, Failure Landscape, and Regulatory Trends

3.1 Layered Architecture and Surfaces

An evidence-based screening system spans four surfaces, shown in Figure 4. A candidate-facing portal collects consent, résumé, and session media. An inference service hosts the perception and learned components. A persistence layer event-sources the session so it can be reconstructed turn by turn. A recruiter-facing application renders the audit. The separation matters because the guarantees this article describes are enforced at boundaries between these surfaces rather than inside any one of them.

Session state is written to an append-only table with a sequence-assigning database trigger. Question plans are deliberately *not* persisted, so a plan cannot be retrospectively edited to match an outcome — a small design decision that removes a whole category of post-hoc rationalisation.

3.2 Failure Landscape

Automated screening fails in ways that conventional information systems do not, because its outputs are judgements about people and its inputs are difficult to verify. Figure 5 organises the failure modes we encountered, and Table 2 states their manifestations and defence implications.

The first two rows are not hypothetical. Section 8 reports an audit of a reference implementation in which both occurred simultaneously: an object detector that never successfully loaded, a luminance heuristic that filled the gap and logged its output as camera-verified detections, and a status string reporting readiness that was never rendered anywhere.

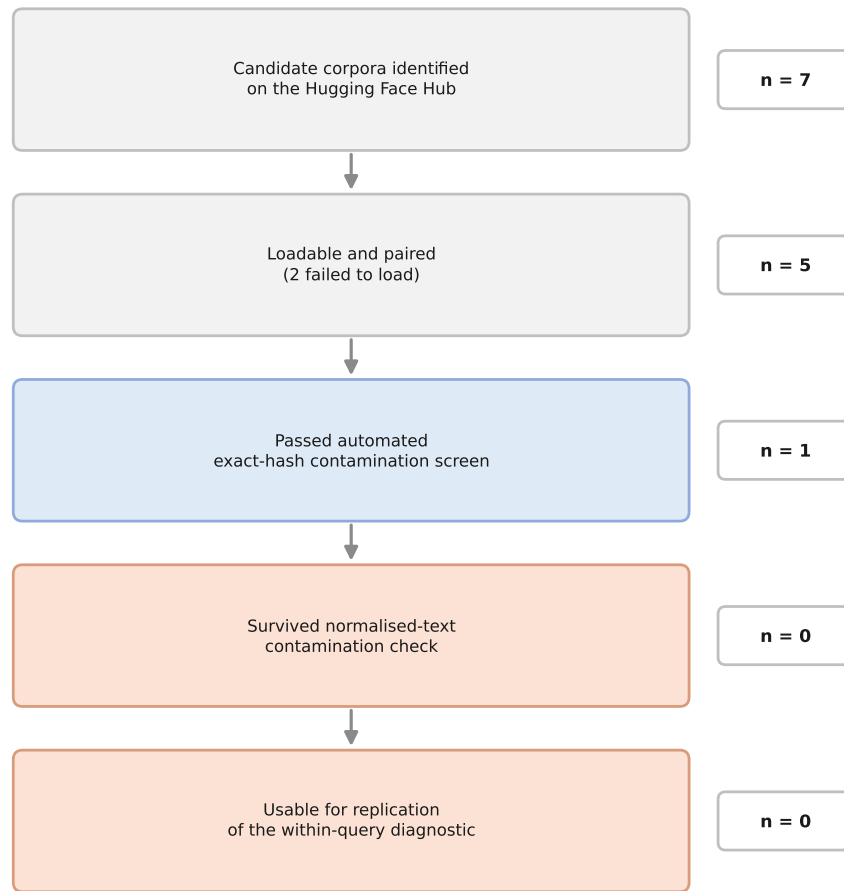


Fig. 3. Replication-corpus screening. Counts are measured, not illustrative: seven candidate corpora were identified, five loaded and carried paired documents, one passed an exact-hash contamination screen, and none survived normalised-text verification (§5).

3.3 Current State and Industry Trends

Table 3 contrasts traditional screening practice with current automated practice and states what each advance costs.

Regulation is converging on explainability. Figure 6 places the four instruments that bear most directly on this system. The Illinois Artificial Intelligence Video Interview Act took effect in 2020 and requires notice and consent for AI analysis of interview video. New York City Local Law 144 requires bias auditing and candidate notification for automated employment decision tools. The European Union Artificial Intelligence Act classifies employment-related systems as high risk, with corresponding transparency and human-oversight obligations. The Illinois Human Rights Act as amended by HB 3773 constrains AI use in employment decisions from 2026.

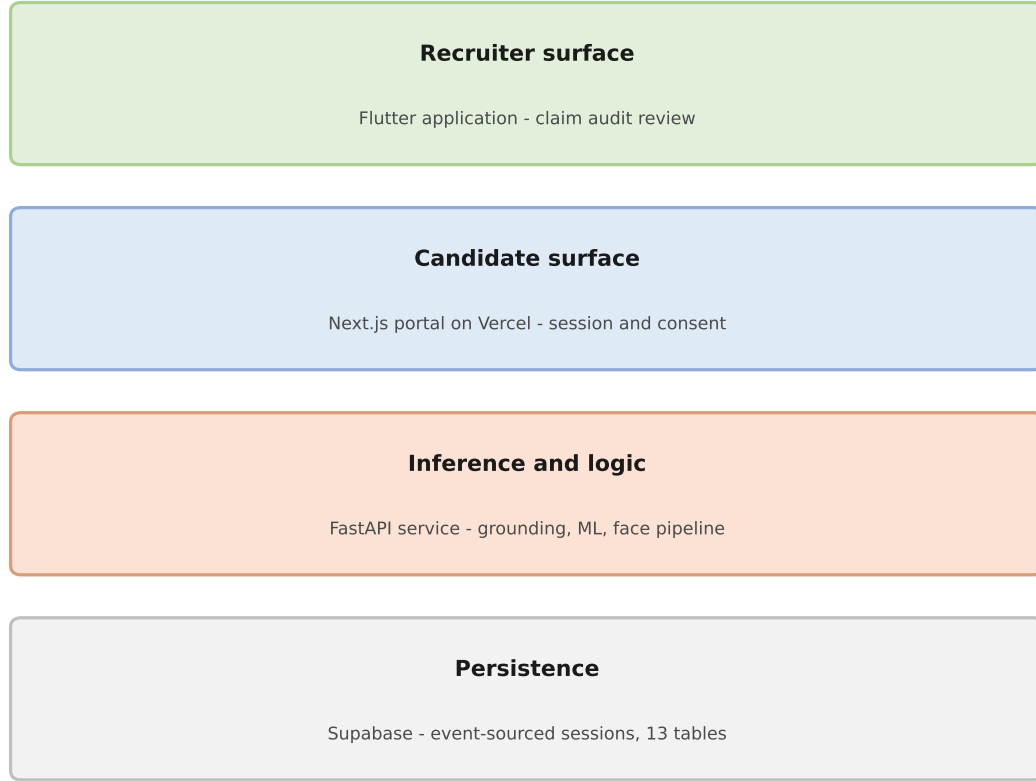


Fig. 4. Layered deployment architecture across the four surfaces.

These instruments share a direction: an automated employment decision must be explicable and contestable. A system architected around a scalar faces a retrofit that its architecture resists, because the explanatory information was discarded at design time. Figure 7 summarises the practices that follow.

3.4 Landscape of Grounding, ML, and DL

Figure 8 organises the technologies examined in the remainder of this article. The division is not arbitrary: grounding governs what may be *asserted*, ML measures properties of *evidence*, and DL supplies *perception*. Assigning a task to the wrong family is itself a failure mode — using a learned model where a rule is correct produces an unauditable decision, and using a rule where measurement is required produces an unvalidated constant.

4 Evidence Grounding as an Enabler

4.1 Grounding-Centric Extraction

The central mechanism of the system is a constraint on what a language model is permitted to produce. Its rule: **a claim's text must be a verbatim substring of the résumé**. The model's role is selection, not generation.

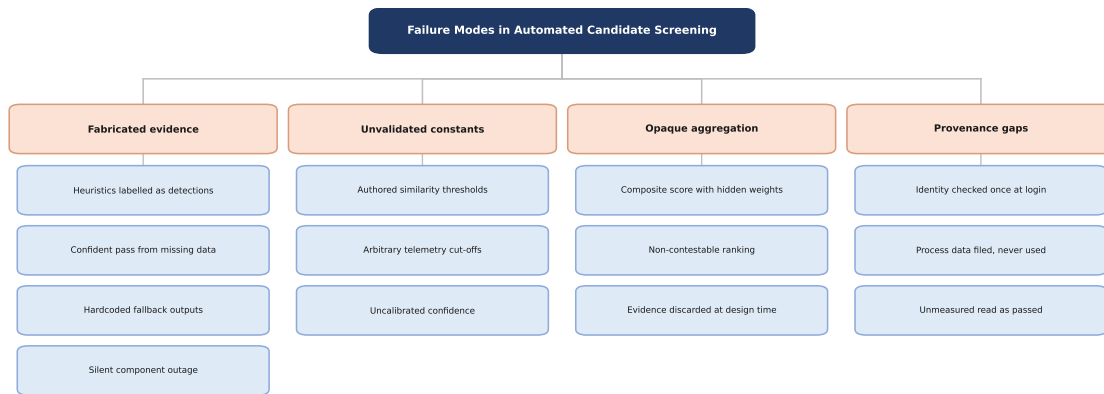


Fig. 5. Failure modes in automated candidate screening. The four families are not independent: fabricated evidence typically originates in an unvalidated constant and is concealed by opaque aggregation.

Table 2. Failure modes in automated screening, their manifestations, and defence implications.

Failure mode	Manifestation	Defence implication
Fabricated evidence	A heuristic proxy is logged under the label of a measurement; a confident pass is returned when the underlying record is absent	Measurements must be structurally distinguishable from inferences; absence must have its own representation
Silent component outage	A model fails to load, a fallback fills the gap, and the status is never surfaced	Health of every perception component must be rendered where a human will see it
Unvalidated constants	A similarity or timing threshold is chosen by reasoning and never measured	Security-adjacent constants require calibration studies with reported error rates
Opaque aggregation	A composite score is produced from hidden weights and cannot be explained or contested	Aggregation should be omitted where the downstream decision is made by a human anyway
Provenance gaps	Identity is verified once at login and assumed thereafter	Provenance must be continuous, and unmeasured intervals must be visible
Instruction injection	Text in a candidate document is interpreted as a directive by a downstream model	Planning must consume typed structures, never raw document text

Naive substring matching is insufficient, and Figure 9 shows the decision pipeline that replaces it. Beyond verbatim containment the gate implements negation rejection, so that “*I have not worked with Kubernetes*” yields no Kubernetes claim despite containing the substring; hedge rejection, so that “*some exposure to Kafka*” is not a Kafka experience claim; contrastive-conjunction splitting, so that “*I used Django, but never in production*” is split rather than truncated; and clause scoping, so that matching occurs within clause boundaries rather than across them.

Enforcement. The grounding module is forbidden from importing the AI module, and an architecture test fails continuous integration if that import appears. The guarantee is therefore a property of the build rather than of developer discipline. Table 4 summarises what this buys.

4.2 Integrity Functions Enabled by Grounding

The last row deserves emphasis. The audit’s verdict set is substantiated, notDemonstrated, contradicted, and notExamined, and the four are deliberately not orderable. notExamined is load-bearing: without it an unprobed claim

Table 3. Comparison of traditional and current automated screening solutions.

Aspect	Traditional	Current	Advances and limitations
Output object	Interviewer notes and a rating	Composite score or ranking	Higher throughput; loses the evidence that justified it
Identity	Visual confirmation in person	One-time check at login	Scales; assumes continuity that remote settings do not guarantee
Process evidence	Observed directly by the interviewer	Captured and filed for retrospective review	Available for fraud investigation; not fed back into the live interview
Consistency	Varies by interviewer	Uniform by construction	Removes one bias source; can entrench another at scale
Explanation	Reconstructed from memory	Feature attributions over a score	Post-hoc; explains the model, not the decision
Contestability	Conversation with the recruiter	Appeal against an opaque number	Formally available, practically difficult
Auditability	Paper trail, incomplete	Logged, but of the score not the evidence	Satisfies retention; not sufficient for a bias audit

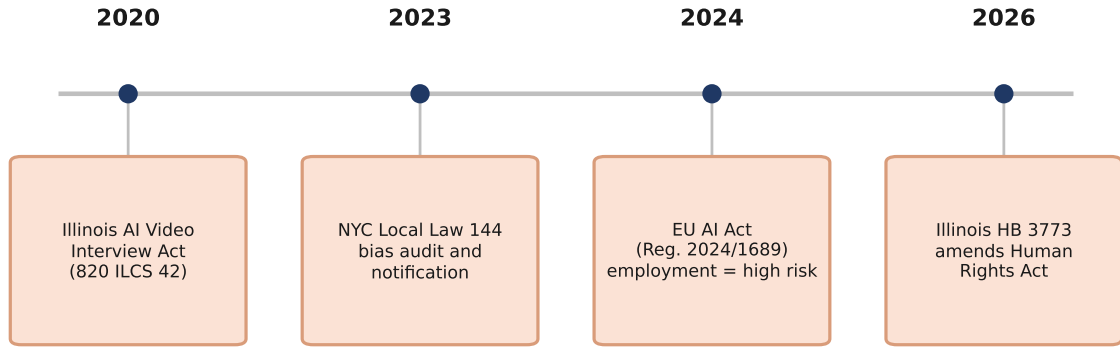


Fig. 6. Regulatory instruments governing automated employment decisions. Dates are statutory effective dates.

is simply absent from the report, and absence reads as acceptance. Making non-measurement explicit is what prevents an audit from being fabricated by omission.

4.3 Measured Behaviour of the Gate

The project's central claim is that the model selects but never authors. The preceding subsections argue this structurally; this one measures it.

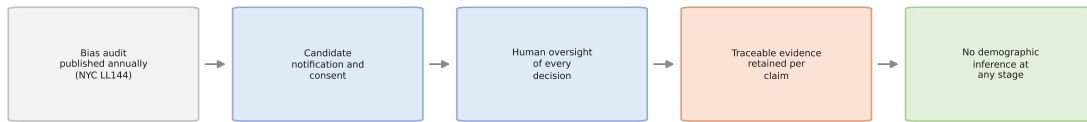


Fig. 7. Compliance practices implied by the instruments in Figure 6.

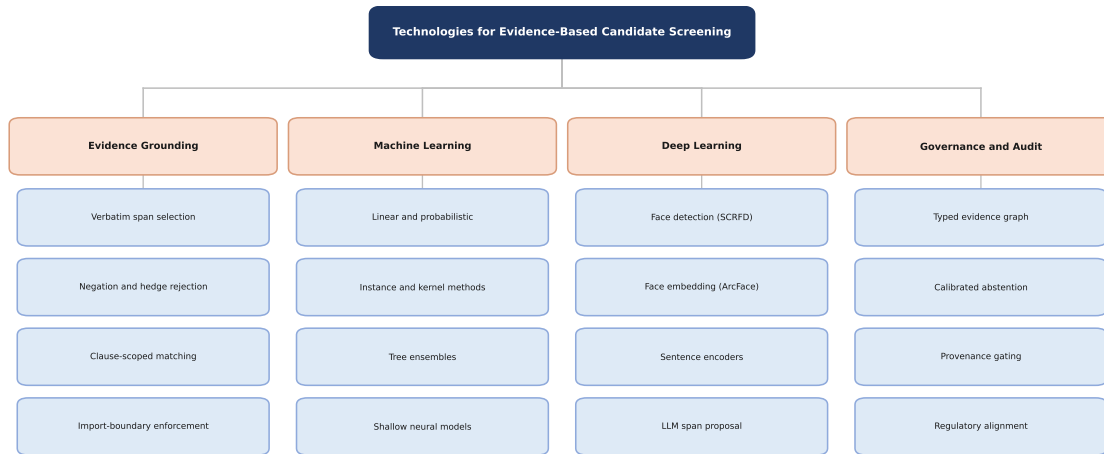


Fig. 8. Taxonomy of technologies for evidence-based candidate screening.

Table 4. Key advantages of evidence grounding.

Advantage	Basis
Verifiable claims	Every claim is a substring with character offsets into the source document
Injection resistance	Résumé text reaches the planner only as typed claims, so an instruction in a document has no field to occupy
Auditable rejection	A discarded claim is discarded by a named rule, not by model preference
Stable evaluation	The gate is deterministic, so its behaviour can be regression tested at corpus scale
No fluency confound	Extraction quality is fidelity to the source, not the plausibility of generated text

An offline deterministic harness drives the real pipeline modules read-only over a corpus of 5,200 synthetic résumés spanning eight fields and four seniority levels. No language-model key is required, so the result is reproducible by anyone. Three quantities are measured across roughly 100,000 trials: the verbatim true-positive rate, the paraphrase rejection rate, and the negation-trap rejection rate. Figure 11 reports the outcome.

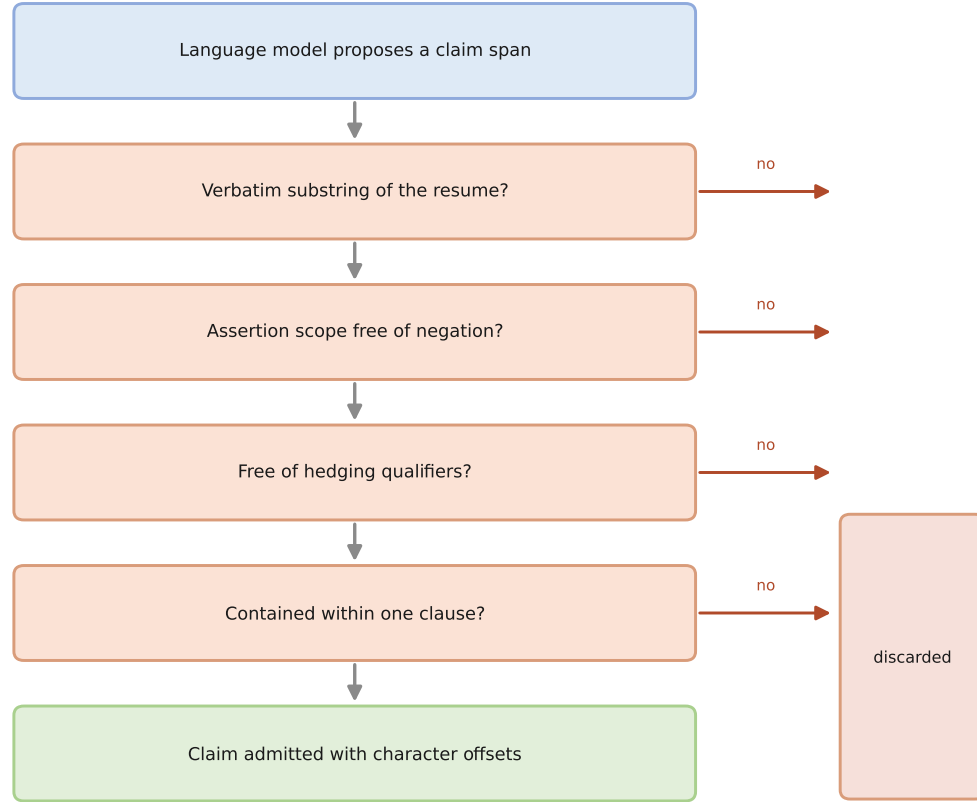


Fig. 9. The grounding gate as a decision pipeline. Any branch marked *no* discards the proposed claim; nothing downstream sees it.

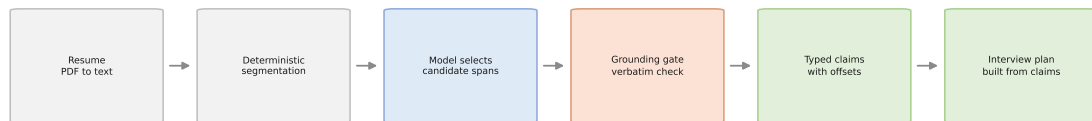


Fig. 10. End-to-end grounded extraction. Segmentation and normalisation remain deterministic; the model contributes span selection only.

The defect this surfaced. The initial run showed a 0.82% false-rejection rate: 425 genuine claims wrongly discarded. Every one contained a period-bearing token — `Node.js`, `React.js`, `asp.net`, `Python 3.9` — and *zero* occurred without one. The clause splitter treated a period, question mark, or exclamation mark as a sentence terminal unconditionally, so “...using JavaScript, Node.js.” split into “...Node” and “js.”, leaving a one-clause claim spanning two clauses where the locator could never find it. The claim was then *silently* rejected: no error, no diagnostic, just an absent claim.

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Table 5. Integrity functions enabled by grounding, and where each is exercised.

Function	Mechanism	Application
Claim provenance	Character offsets retained from extraction through to the audit	A reviewer clicks a claim and lands on the candidate’s own words
Prompt-injection defence	Planner consumes typed claims only	A résumé containing an embedded instruction has no channel to the planner
Assertion-scope fidelity	Negation and hedge detection inside clause boundaries	Denials and qualifications are not converted into claims
Quote fidelity in speech	Spoken-turn quotes must match the transcript verbatim, including disfluency	An interviewer cannot attribute tidied speech to a candidate
Non-fabrication of coverage	Unexamined claims carry an explicit verdict	Absence of evidence is reported, not omitted

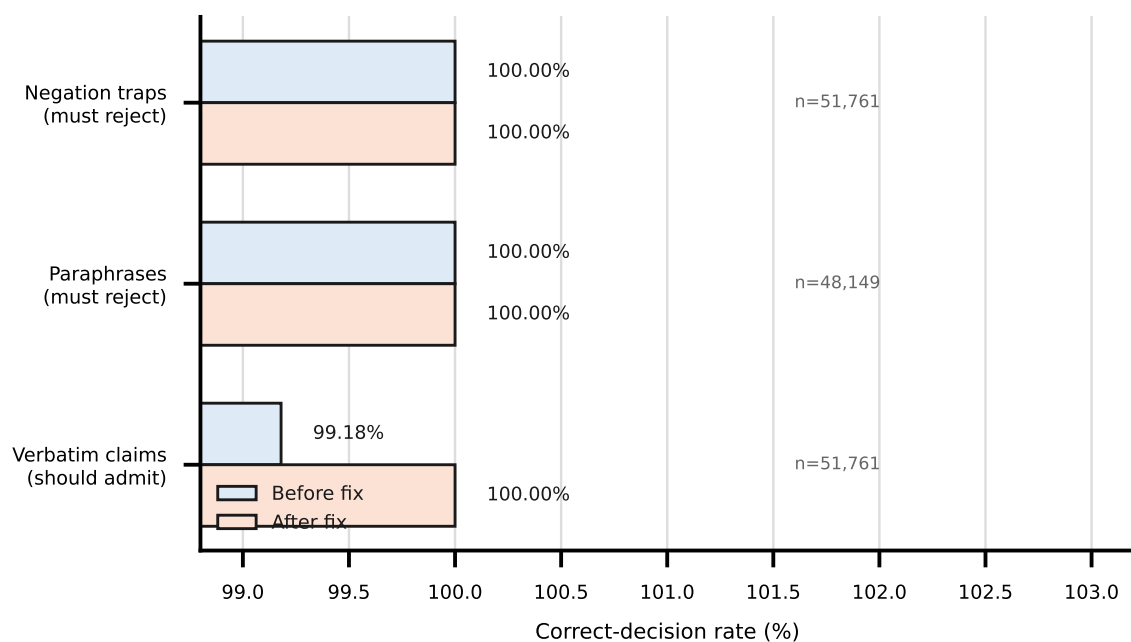


Fig. 11. Grounding-gate evaluation before and after the clause-splitter fix described below. Trial counts are shown at right.

The fix constrains a terminal to count only when followed by whitespace or end-of-string. A real sentence terminal always is; the period between two word characters never is. Crucially, keeping dotted tokens whole makes clauses *longer*, which widens negation scope rather than narrowing it, so the fix cannot weaken the guard — and the measured before/after confirms it, with both rejection rates unchanged at exactly 100%.

This is a failure mode worth naming: the bug was *silent*. Nothing crashed and no test failed; claims simply went missing. Only a corpus-scale evaluation with an explicit true-positive metric could surface it, which is an argument for measuring the properties you claim rather than only testing the paths you wrote.

Table 6. Future directions for grounded extraction and their expected impact.

Direction	Expected impact
Compositional claims	Extracting technology, artefact, metric, and agency as typed relations rather than as a single span
Multilingual scope	Negation and hedge detection currently assume English clause structure
Discontiguous spans	Some true claims are expressible only across non-contiguous text; a discontiguous grounding relation would admit them without loosening the gate
Speech grounding	Applying the constraint to spoken turns requires treating the transcript as the source of truth, with its own error rate

Honest scope. The corpus is synthetic and produced by a single generator, so 100% extraction coverage is an artifact of that format and is *not* claimed as a general result. The defensible claim is narrower and concerns the safety mechanism: the gate rejected 100% of model-style paraphrases and negation traps across roughly 100,000 trials while admitting 100% of verbatim claims.

4.4 Open Challenges in Grounded Extraction

The gate’s principal cost is recall on claims a résumé expresses across a sentence boundary. Tightening that without loosening the safety property is the main open problem in this component.

5 Machine Learning Techniques for Candidate Screening

5.1 Taxonomy of ML Paradigms in Screening

Machine learning enters an evidence-based screening system in exactly one place: measuring properties of the evidence a session has accumulated. It does not decide, and it is not trained on hiring outcomes. That restriction is what makes the remainder of this section a measurement problem rather than a prediction problem about people. Table 7 states the paradigms and where each is admissible.

5.2 Representative Models and Use Cases

Figure 12 shows the families compared, and Figure 13 places them against the evolution of screening automation. Table 8 summarises how the literature has applied these families.

5.3 A Nine-Model, Six-Family Comparison

We compare nine classifiers spanning six model families on a public résumé-posting fit benchmark [1] of 7,910 balanced pairs drawn from 643 résumés and 351 postings. Every experiment uses ten-fold GroupKFold grouped by job posting, so each fold’s test postings are absent from its training data. Two quantities are fitted inside each fold on training rows only — the IDF corpus statistics and the feature scaler — because fitting either over the pooled corpus would leak the test fold’s term distribution into training. Identical fold assignments are reused across every model, so all comparisons are paired.

Table 7. ML paradigms in candidate screening and their admissibility under an evidence-audit design.

Paradigm	Typical screening use	Admissible here?	Reason
Supervised, outcome-labelled	Predict hire / no-hire from historical decisions	No	Inherits whatever the historical decisions encoded; this is the failure the 2018 Amazon case made famous
Supervised, evidence-labelled	Predict whether a session gathered enough material to judge a claim	Yes	Target is a property of the interview, not of the person
Relational / matching	Score a résumé against a posting	With care	Pooled metrics conflate matching with candidate ranking (§5.4)
Unsupervised	Cluster claims, deduplicate skills	Yes	Structural, no verdict produced
Self-supervised	Capture-quality prediction from embedding stability	Yes	No human annotation, no person-level label
Reinforcement learning	Adaptive question selection	Offline only	Online exploration means deliberately asking a real candidate a question believed to be poor

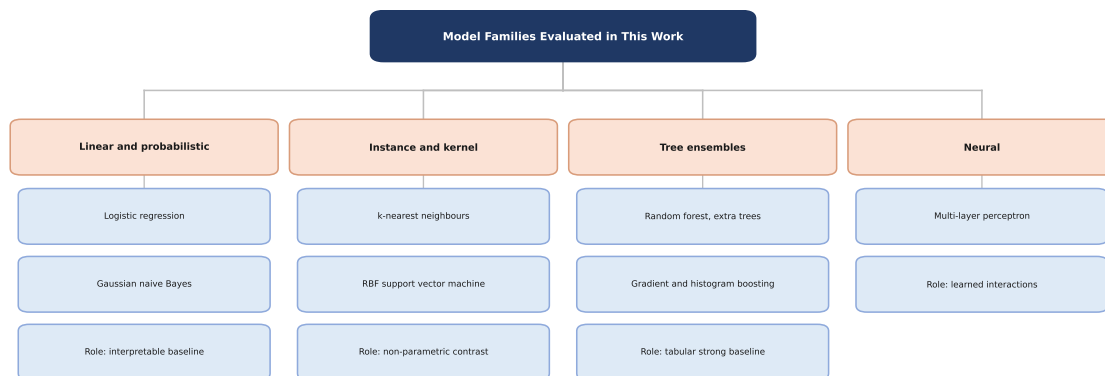


Fig. 12. Model families evaluated in this work and the role each plays in the comparison.



Fig. 13. Evolution of screening automation, from lexical matching to grounded evidence audit.

Models run at fixed modest hyper-parameters with no per-model tuning budget; §5.6 tests whether that choice matters. Scale-sensitive models receive standardised features and tree ensembles receive raw features, since standardising a scale-invariant model would compare preprocessing rather than algorithms.

The nine span 0.031 AUROC from best to worst, and every model’s fold-to-fold standard deviation is of the same order as that entire spread. The Friedman test rejects the hypothesis that all nine perform identically ($\chi^2 = 38.35$, $p = 6.5 \times 10^{-6}$), so the models are not interchangeable — but the post-hoc picture is what governs a ranking claim.

Table 8. Methodological studies bearing on model comparison in screening-like tasks, and the role each plays in this article’s protocol.

Refs	Technique / contribution	Focus	Role in this work
[2]	Friedman test with post-hoc procedures	Comparing many classifiers	Adopted as the significance protocol
[11]	Corrected variance for overlapping folds	Generalisation-error inference	Motivates non-parametric paired tests
[4]	Reproduction under controlled protocol	Neural recommenders vs. heuristics	Template for a null result that is informative
[10]	Evaluation-practice audit	Deep metric learning	Precedent for attributing gains to protocol
[5]	Randomly regenerated splits	Standard-split fragility	Motivates ten grouped folds over one split
[9]	Eight-type leakage taxonomy	ML-based science	One category quantified in §5.5
[6]	Trees vs. deep nets, 45 datasets	Tabular benchmarks	Context for family-level spread
[12]	Reference implementations	Toolkit	All nine models are its estimators

Datasets, baselines, and evaluation metrics differ across the cited papers; the table records each work’s role, not a performance comparison.

Table 9. Nine classifiers on résumé–posting fit, ten-fold GroupKFold by posting. Efficiency columns are per-fold means.

Model	AUROC	Accuracy	Macro F1	ECE	Fit (s)	Latency (μ s/row)	Size (KB)
MLP	0.696 $\pm$ 0.031	0.634 $\pm$ 0.021	0.658 $\pm$ 0.038	0.057	0.44	0.5	74.7
Gradient boosting	0.689 $\pm$ 0.028	0.636 $\pm$ 0.024	0.655 $\pm$ 0.041	0.047	2.44	1.9	393.3
Random forest	0.688 $\pm$ 0.030	0.628 $\pm$ 0.026	0.647 $\pm$ 0.042	0.047	0.59	88.2	24,732.6
SVM (RBF)	0.686 $\pm$ 0.028	0.635 $\pm$ 0.023	0.651 $\pm$ 0.039	0.053	4.64	195.0	526.8
Extra trees	0.685 $\pm$ 0.030	0.632 $\pm$ 0.026	0.641 $\pm$ 0.044	0.046	0.39	84.9	63,415.9
k -NN	0.680 $\pm$ 0.031	0.628 $\pm$ 0.029	0.639 $\pm$ 0.041	0.057	0.01	35.5	1,272.8
Hist. gradient boosting	0.679 $\pm$ 0.029	0.626 $\pm$ 0.020	0.643 $\pm$ 0.037	0.057	0.41	7.3	591.8
Gaussian NB	0.667 $\pm$ 0.033	0.620 $\pm$ 0.027	0.633 $\pm$ 0.051	0.117	0.00	0.3	1.1
Logistic regression	0.665 $\pm$ 0.022	0.616 $\pm$ 0.020	0.637 $\pm$ 0.042	0.052	0.01	0.2	0.9
<i>Fixed-threshold baseline</i>	—	<i>0.513 $\pm$ 0.028</i>	—	—	—	—	—

Comparing the leader against each other model with a paired Wilcoxon signed-rank test over the ten folds and applying Holm–Bonferroni correction, the leading model is **statistically indistinguishable from all four tree ensembles**, separating only from the linear, probabilistic, instance-based, and kernel baselines, where the largest effect is three AUROC points.

A study reporting only the ranked table would license the claim “the MLP is the best model for résumé–job fit.” The significance analysis does not support it. Note also the non-monotonicity across metrics: the MLP leads on AUROC, gradient boosting on accuracy, extra trees on calibration. Which model “wins” depends on which column is placed first.

Establishing that the protocol has power. A comparison that finds no difference is uninformative unless the protocol can be shown to detect a difference when one exists. We therefore constructed a synthetic control with a pre-specified generative process — twelve features, eight carrying linear weights, one interaction term, one quadratic term, four weightless, and a 7% label-flip rate imposing a Bayes-optimal accuracy of 0.925 — and ran it through the identical protocol. There the same nine models separate decisively ($\chi^2 = 55.42$, $p = 3.7 \times 10^{-9}$), with the best recovering 94.9% of achievable accuracy.

Figure 14 places the two side by side. *The synthetic task is not a résumé task and its accuracy figures are not screening performance.* It exists to characterise the protocol’s power, and every artifact it emits is flagged as synthetic. Its role is that of a positive control in an assay: an instrument that reports nothing must be shown capable of reporting something.

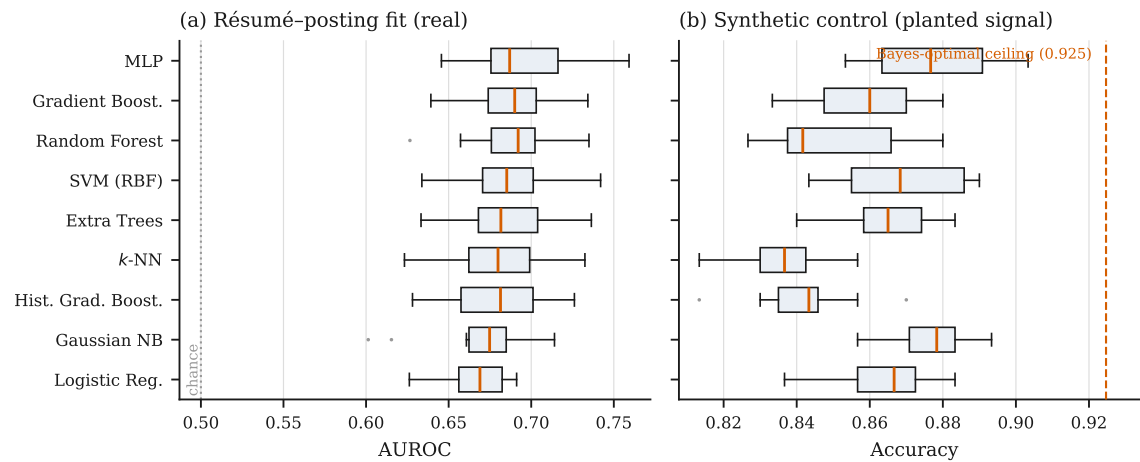


Fig. 14. Per-fold distributions over ten grouped folds. **(a)** The nine models on the real screening task; the boxes overlap almost completely. **(b)** The same nine models, same folds, same code path, on the synthetic control, against its Bayes-optimal ceiling. Models appear in the same order in both panels. The contrast is the power analysis: an instrument reporting no difference in (a) is shown in (b) to be capable of reporting one.

The comparison licenses one inference — when learnable signal is present the protocol resolves it clearly, and on the résumé task it does not — so the null is a property of the task, not a limitation of the study.

5.4 Separating Matching from Candidate Quality

A pooled metric rewards two distinct capabilities and cannot separate them. A model that knows which *résumés* tend to be labelled fit will score well, and so will a model that knows which *postings* a given *résumé* fits. Only the second is matching.

Varying the representation while holding the protocol fixed localises where performance comes from. A probe given only the résumé embedding — which cannot be matching, since it never sees the thing being matched against — reaches 0.660 AUROC, 87% of the best model’s score and statistically indistinguishable from a single cosine-similarity feature. Its mirror, given only the posting, reaches 0.546, barely above chance. The best representation overall is plain concatenation of the two embeddings at 0.755, which is to say that the representation most enabling résumé recognition scores highest.

Figure 15 shows the ordering. That evidence pointed us toward a strong conclusion — that the benchmark’s labels were not relational at all — and a direct test refuted it. Decomposing label variance shows **80.6% is within-résumé**: the same résumé labelled fit for one posting and not for another. To measure whether models access that variance we restrict evaluation to pairs of rows sharing a résumé, one positive and one negative, and ask whether the model ranks the positive posting higher. Because both items share a résumé, every résumé-level property cancels exactly, and chance is 0.5 by construction.

Models do match. Both representations rank within-résumé pairs well above chance by every test applied. A model given one résumé and two postings picks the fitting one roughly 63% of the time.

But the pooled gain does not come from matching. Moving from the ten engineered features to the 1536-dimensional interaction encoding raises between-résumé ranking by +0.047 and pooled AUROC by +0.041, while

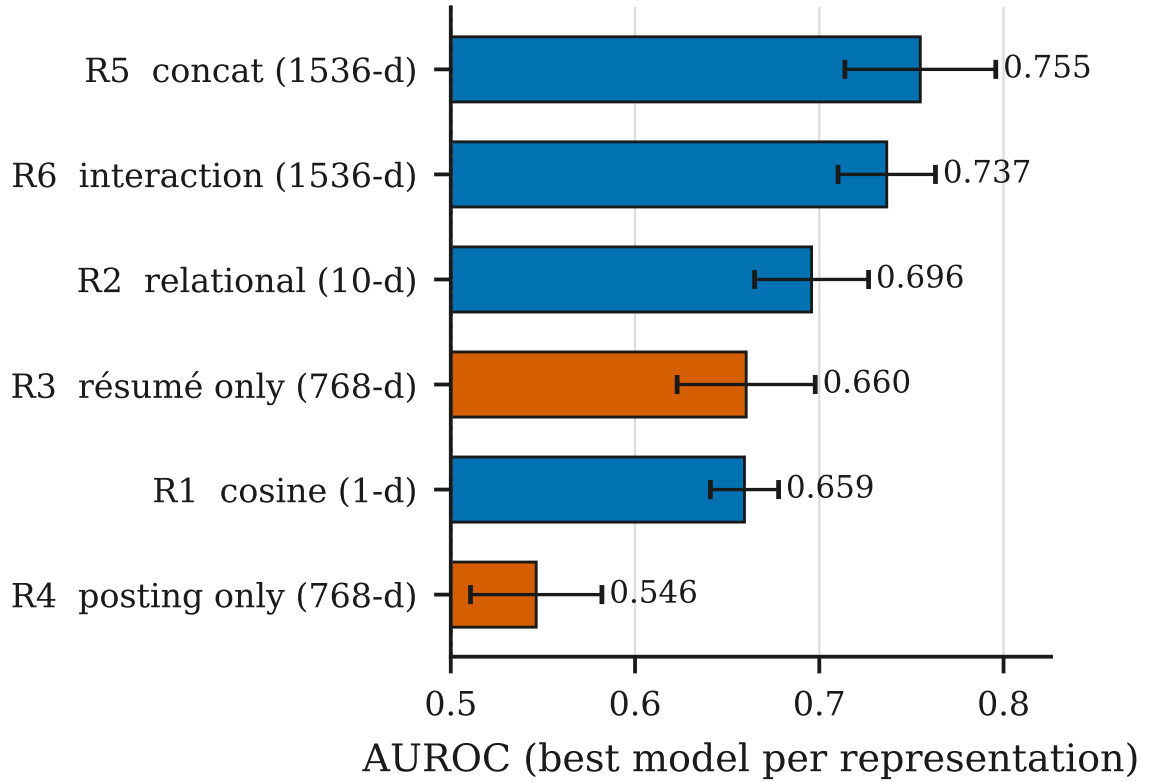


Fig. 15. Best AUROC per representation, identical folds throughout. The highlighted bars are one-sided probes that cannot be performing matching: one never sees the posting, the other never sees the résumé.

Table 10. Pairwise ranking accuracy. Random forest, identical folds, chance = 0.5, 4,112 within-résumé pairs.

Representation	Between	Within	Binomial p
Relational (10-d)	0.6905	0.6333	3.8×10^{-66}
Interaction (1536-d)	0.7372	0.6240	2.0×10^{-57}

within-résumé matching changes by -0.009 , which a paired Wilcoxon test over the ten folds does not distinguish from zero ($p = 0.445$). We state this as *the gain does not transfer*, not as *matching degrades*: the data support the absence of an improvement, not the presence of a loss.

This is the section’s central measurement. The representation change a leaderboard would record as a clear improvement bought a better ordering of résumés against each other and left the model’s ability to tell which posting suits a candidate where it was. Nothing in the pooled metric reveals which of the two capabilities moved.

The diagnostic generalises. Nothing in the construction is specific to résumés. Any benchmark whose instances are (entity, item) pairs with recurring entities — candidate–job, patient–treatment, user–item, student–question, query–document — admits the same test: condition on the entity, rank within it, compare against the pooled number. It needs no extra labels and one pass over predictions already computed. Where the two diverge, the pooled number is

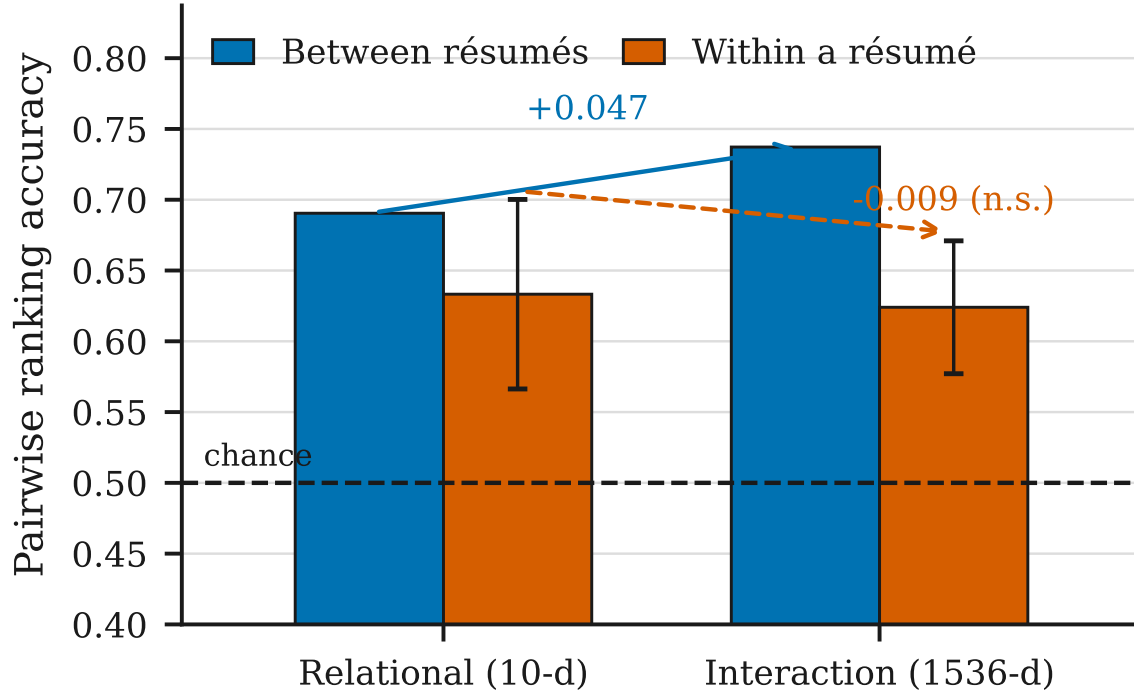


Fig. 16. Pairwise ranking accuracy, chance = 0.5. Moving to the richer representation raises between-résumé ranking by 0.047 while leaving within-résumé matching unchanged (-0.009 , $p = 0.445$).

Table 11. Split-scheme comparison. Identical features and model throughout.

Scheme	Held out from training	AUROC
S1 k -fold, no grouping	nothing	0.7137 ± 0.016
S3 GroupKFold by résumé	every test résumé	0.7032 ± 0.031
S2 GroupKFold by posting	every test posting	0.6894 ± 0.016

measuring something other than the advertised capability. For algorithmic hiring specifically, this is the measurement that distinguishes a system that ranks matches from one that ranks people — precisely the target substitution Raghavan et al. [14] warn carries the ethical and legal weight.

5.5 Robustness of the Comparison

Two objections bear on the interpretation, and both are testable.

Does the protocol still permit résumé leakage? Table 11 varies only the splitting scheme. The ungrouped scheme a conventional bake-off would use scores 0.024 AUROC above the posting-grouped protocol we adopt, so the naive protocol inflates, modestly. But closing the résumé channel does not collapse performance — it raises it. We expected the résumé-grouped scheme to fall sharply if the model’s apparent skill were résumé recognition; it scores 0.014 *above* the posting-grouped scheme. A strong-memorisation account predicts the opposite sign and is refuted. The ordering

instead tracks which side of the pair is novel, and generalising to a new posting is the harder problem — which matters in deployment, where every new requisition presents exactly that condition.

Did binarising a three-class label destroy the signal? The cleanest available contrast, Good Fit against No Fit with the ambiguous middle class removed, does *not* exceed the binarisation used (0.6810 vs. 0.6894), and its fold variance triples. Binarisation is not concealing signal.

5.6 Does Tuning Change the Conclusion?

The claim that nine models are statistically indistinguishable is vulnerable to the objection that they were left untuned. We tested it with a design that could have refuted the claim: a nested cross-validation whose outer loop is the same ten folds and whose inner three-fold loop selects among eight configurations per model — an identical budget for every model, since an unequal budget compares search effort rather than algorithms.

Tuning rescues no model. The largest gain across all nine is +0.0077 AUROC and three models are marginally worse tuned than untuned, which is ordinary nested-CV selection noise. The best-to-worst spread is essentially unchanged (0.0311 $\rightarrow$ 0.0317), but the significance structure *contracts*: after Holm correction the tuned leader separates from only two of eight rivals where the untuned leader separated from four. The two models that had been significantly behind and gained most from tuning are no longer distinguishable from the leader. Equal-budget tuning therefore leaves the ranking substantively unchanged and weakens rather than strengthens the case that any model is best.

5.7 Architectural Integration of ML

Figure 17 shows placement. Two considerations drive it. Scoring that must survive a network outage runs on-device, which is why the evidence sufficiency model ships as exported coefficients rather than as a service call. Components requiring model weights too large to ship run in the service. Training and calibration are strictly offline and gated: the export refuses to write an artifact unless held-out AUROC, calibration error, and Brier score all clear stated bars, so a bad run produces no file rather than a quietly bad one.

Efficiency is decoupled from accuracy. Across the nine models, inference latency spans 0.2 to 195.0 μ s per row — a factor of roughly 1,000 — and serialised size spans 0.9 KB to 63.4 MB, a factor of about 70,000. Predictive quality spans 0.031 AUROC. Figure 18 shows the frontier, which is nearly flat. For an interactive deployment the rational choice is therefore the cheapest model that is not significantly worse, which here is a kilobyte of logistic-regression coefficients rather than a 64-megabyte forest.

More data would not close the gap. A learning curve over the same folds rises from 0.6514 AUROC at a tenth of the training data to 0.6878 at all of it. Quadrupling the data from a quarter to full buys 0.0204; doubling from half to full buys 0.0076. The curve is flattening well below any useful operating point (Figure 19), so the limitation is not sample size and collecting more rows of this kind would not resolve it.

5.8 Open Challenges and Research Directions

The first row is a finding rather than a limitation of effort. A benchmark whose most-downloaded derivatives regenerate its labels by model and re-pair its documents combinatorially is a benchmark whose apparent diversity is illusory: work reporting evaluation across “several résumé-fit datasets” may be evaluating repeatedly on one. We note also that an exact-hash contamination check passed the contaminated corpus as clean, and only normalised matching exposed

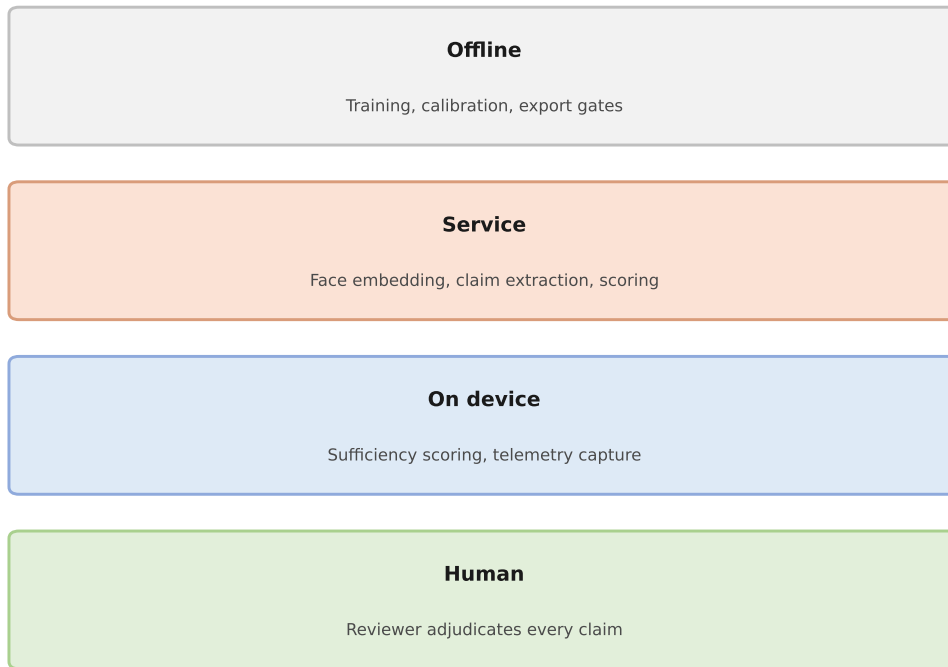


Fig. 17. Where each learned component runs. The human tier is not a fallback; it is the only tier that produces a decision about a person.

Table 12. Challenges in ML for evidence-based screening.

Challenge	Impact and direction
No independent corpus	Of seven public résumé-fit corpora screened, the largest shares 100% of its documents with the corpus under study and the independent ones pair each résumé with exactly one posting, making the within-query statistic undefined
Undocumented provenance	The benchmark carries no description, licence, citation, or labelling procedure, so what a classifier trained on it has learned cannot be stated
Evidence labels are costly	Annotating evidence sufficiency requires session replay and multiple annotators; agreement is the model's performance ceiling and must be reported
Representation ceiling	Changing representation moves AUROC about twice as far as changing algorithm; effort is better spent there

it. Contamination audits should normalise before hashing and compare entity sets rather than only pair sets, since re-pairing the same documents preserves document overlap while destroying pair overlap.

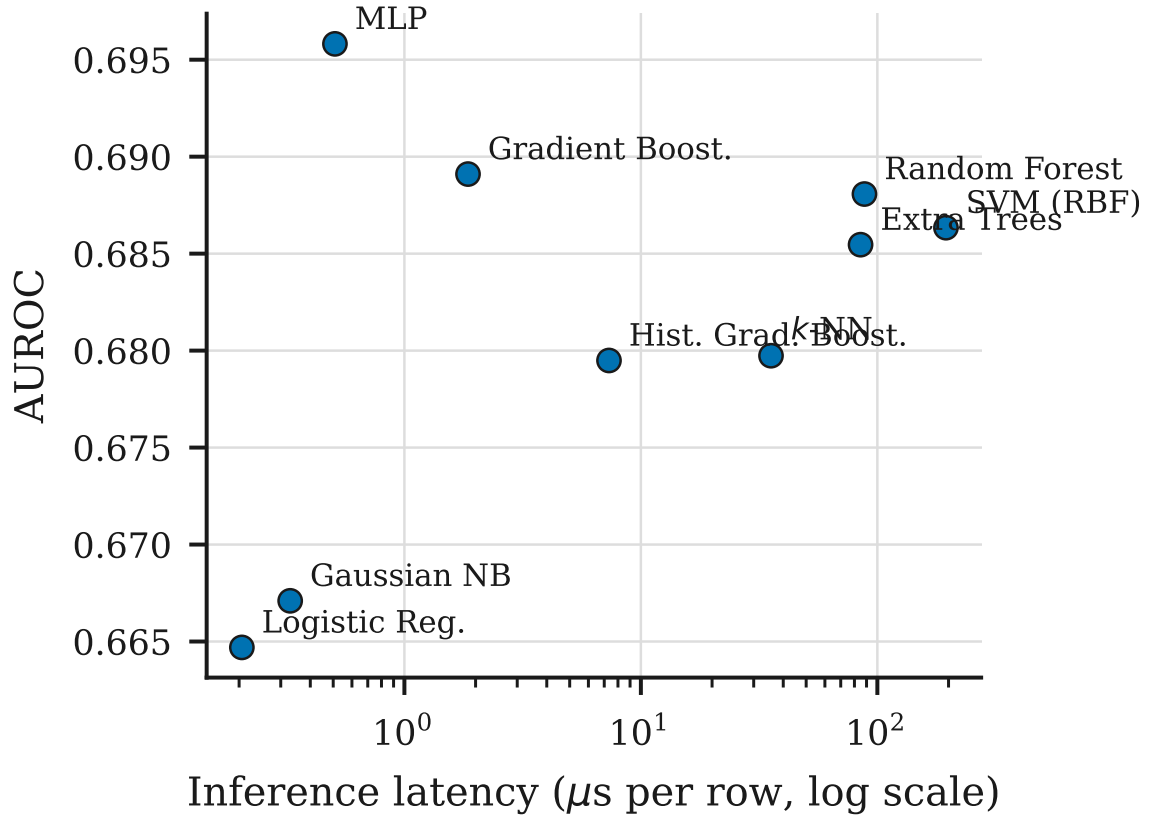


Fig. 18. AUROC against inference latency (log scale). Three orders of magnitude of latency buy 0.031 AUROC, most of which is not statistically resolvable.

6 Deep Learning for Identity and Extraction

6.1 DL Architectures

Two deep architectures carry load in this system, and neither is trained here. Face detection uses SCRFD [7], whose contribution is redistributing computation between backbone, neck, and head to detect small faces efficiently. Face embedding uses ArcFace [3], whose additive angular margin loss produces highly discriminative 512-dimensional embeddings. Both are consumed as pretrained components.

Why the encoder is frozen. Retraining face recognition on a project-scale dataset would be strictly worse than the pretrained model and would introduce demographic bias the project cannot audit. The research question is not the encoder; it is what is done with the embedding stream downstream. Relatedly, the vendor model pack ships an age and gender classifier, and the loader specifies an allow-list that never loads it — a single argument a reviewer can verify, rather than a policy statement.

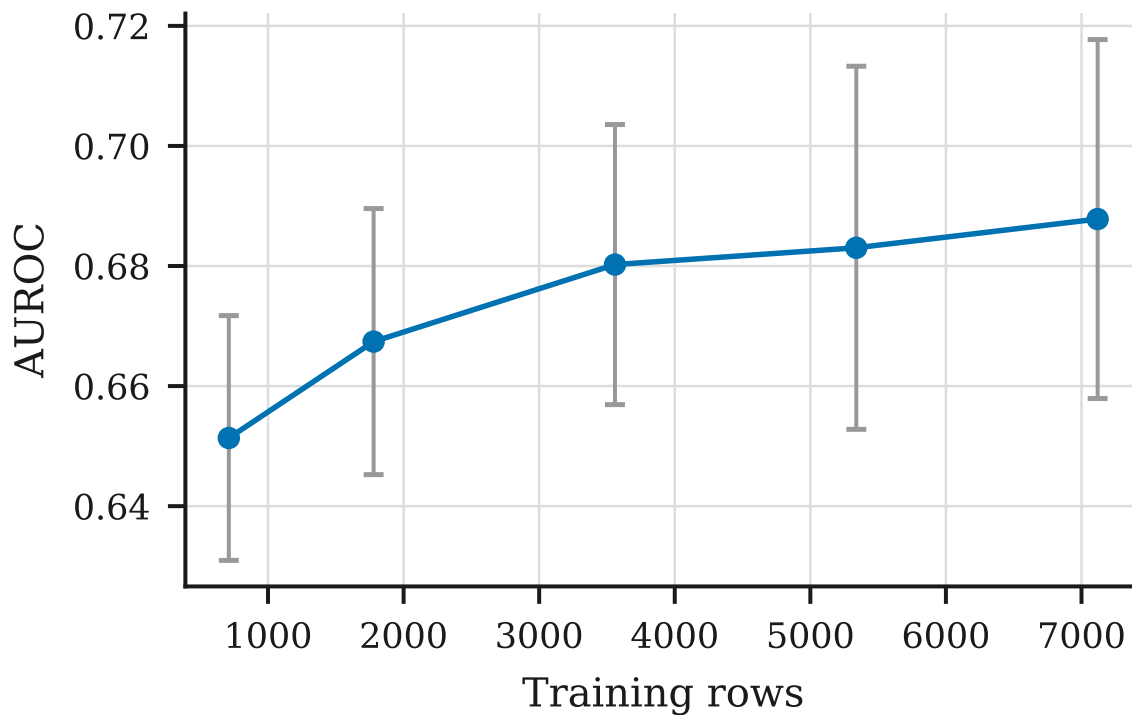


Fig. 19. Learning curve over the same ten folds. The curve flattens well below any useful operating point.

Table 13. Deep architectures in the pipeline: advantages, limitations, and use.

Architecture	Advantage	Limitation	Use here
SCRFD detector	Efficient across compute regimes; strong on small faces	Detection only; no identity	Locating faces in webcam frames
ArcFace embedding	Angular margin yields separable identity geometry	Threshold left to the integrator	512-d embedding for identity continuity
Sentence encoder	Dense semantic comparison of free text	Similarity is not entailment	Claim clustering and relational features
Language model (gated)	Fluent span proposal over messy documents	Will author text if unconstrained	Proposes spans; the gate decides

6.2 Threshold Calibration and Abstention

The system originally compared embeddings against an authored similarity threshold of 0.50 — a value arrived at by reasoning, documented honestly, and flagged in its own source comment as uncalibrated. We replaced it with a measured one. Using the Labeled Faces in the Wild benchmark [8], an equal-error-rate sweep on 2,200 training pairs and evaluation on 1,000 held-out pairs yields the results in Table 14. No weights are trained; a single scalar is fitted.

This is the article’s most consequential single finding. The prior threshold was not arbitrary; it was reasoned, documented, and plausible. It was also wrong in a way nobody could have known without measuring: it would have falsely rejected 41.4% of genuine identity matches. In deployment, roughly two in five legitimate re-verification checks

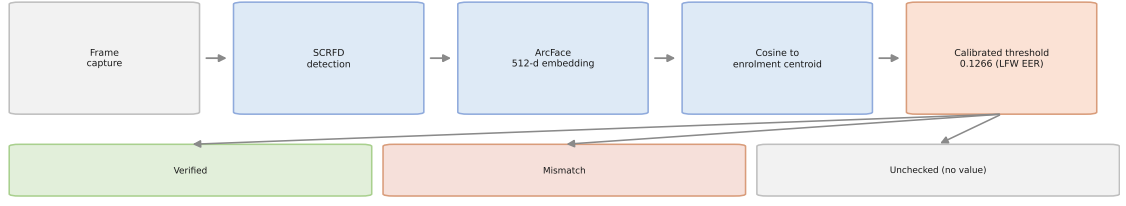


Fig. 20. Identity pipeline. The result type is three-valued; *Unchecked* carries a reason and no similarity value at all.

Table 14. Face-verification threshold before and after calibration on LFW.

Threshold	FAR	FRR	Accuracy	AUC
0.1266 (calibrated)	0.030	0.034	0.968	0.9785
0.50 (authored)	0.000	0.414	0.793	—

would have failed — and because the system correctly refuses to treat an unmeasured check as a pass, those candidates would have accumulated integrity flags for being themselves. Calibration reduced false rejection by a factor of twelve at a false-acceptance cost of 3.0%. The lesson generalises beyond this component: in security-adjacent systems, *reasoned* and *correct* are different properties, and only one of them can be demonstrated.

Abstention as a hard gate. Figure 20 shows the three-valued result type. *Unchecked* carries a reason and has *no similarity field at all*. It is not similarity 0.0, not null, not a default — the value does not exist, so no downstream comparison can read a failed measurement as a low-but-real one and no serialisation can emit a number that was never measured. Selective prediction of this kind is a hard constraint rather than a soft preference, in the spirit of distribution-free abstention [15]; a model that *usually* respects a safety constraint is a model that sometimes does not.

Stated limitation. LFW comprises public-figure photographs, not this system’s candidate population. The calibrated value is a defensible general threshold, not one validated on real candidates, and population-specific recalibration requires consented candidate data we have not collected.

6.3 Deployment Challenges

7 Synergistic Integration of Grounding, ML, and DL

7.1 Motivation for Integration

Each family alone leaves a gap. Grounding guarantees that claims are the candidate’s own words but says nothing about whether a session gathered enough evidence to judge them. ML measures evidence sufficiency but, without grounding, would measure the sufficiency of text a model may have invented. DL supplies identity continuity but produces no claim and no verdict. The combination closes a loop that none closes alone, and Figure 21 shows its shape — with the human reviewer at the centre, because the loop deliberately terminates in a person rather than in an aggregate.

7.2 Evaluated Benefits and Impact

Table 16 collects the measured effect of integration. Three observations follow. First, the largest single improvement came not from a model but from measuring a constant. Second, two rows read “unmeasured” before — the property

Table 15. Challenges, impacts, and mitigations for DL components.

Challenge	Impact	Mitigation
Population shift	Benchmark-calibrated thresholds may not transfer	Report the mismatch; recalibrate under consent
Capture quality	Poor lighting widens within-session variance	Abstain rather than mismatch
Silent model failure	A failed load can be masked by a fallback	Surface component health in the UI; never substitute a heuristic
Embedding privacy	Face embeddings are invertible to recognisable faces	Retain on-device, encrypted, with a deletion window
Demographic inference	Bundled classifiers infer protected attributes	Allow-list modules; never load them

Table 16. Key metrics before and after integration. Every row is a measured value from this work, not an estimate.

Metric	Before	After	Basis
Identity false-rejection rate	0.414 (authored threshold 0.50)	0.034 (calibrated 0.1266)	LFW, 1,000 held-out pairs
Paraphrase leakage	Unmeasured	0 of 48,149	Offline gate harness
Negation-trap leakage	Unmeasured	0 of 51,761	Offline gate harness
Verbatim admission	99.18% (425 wrongly rejected)	100.00%	Clause-splitter fix
Model-selection basis	Ranked table	Friedman + Holm-corrected pairwise	Ten grouped folds
Matching vs. ranking	Conflated in one metric	Separated, 0.633 within-résumé	4,112 within-résumé pairs
Composite score	Present in comparable systems	Absent by construction	Vocabulary ban fails the build

was asserted rather than tested, which is the condition this article argues is most common and least noticed. Third, the last row is a capability *removed*: the system cannot emit a composite score, and that absence is enforced by a test.

7.3 Architecture as Guarantee

The recurring engineering move across all three families is converting a *policy* into a *structure*.

Each row converts a claim that requires trust into one that can be checked in seconds by someone who does not trust us. That is the property we would most like other screening systems to adopt, independent of any of the specific technologies in this article.

7.4 Integration Challenges and Design Considerations

7.5 Standards and Regulatory Compliance

The mapping in Table 19 is possible only because the system declines to produce an aggregate. A bias audit of a composite score must explain the score; a bias audit of a claim audit examines the distribution of verdicts and the evidence attached to each, which is a smaller and better-posed question.

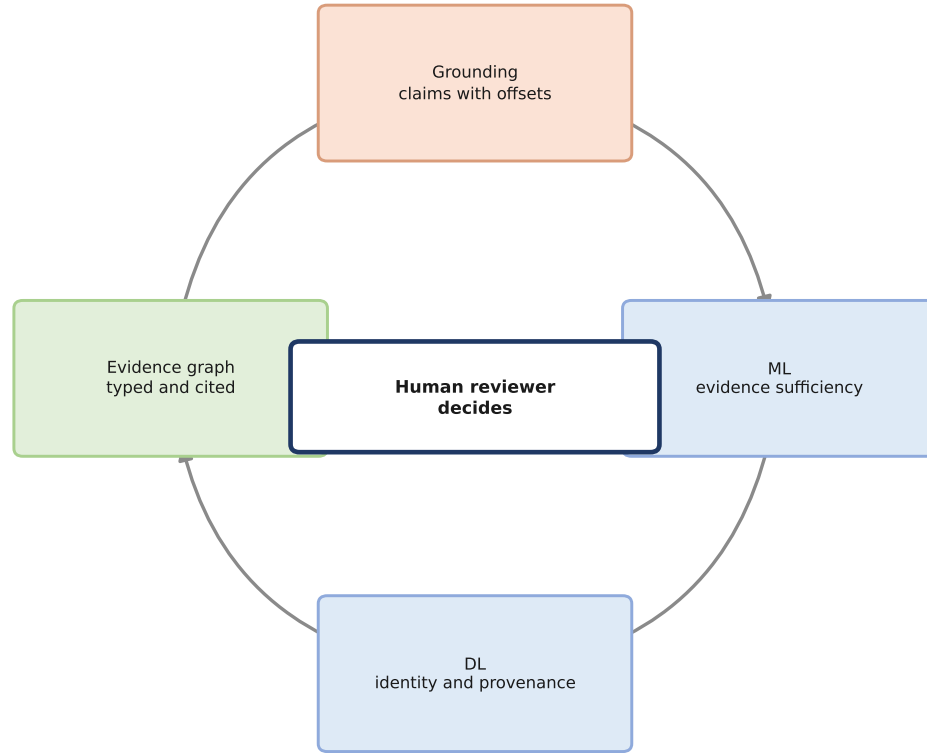


Fig. 21. The integrated loop. Every arrow carries typed, cited evidence; the reviewer is the only component that produces a decision about a person.

8 Deployment Experiences and Lessons

8.1 Use Cases of Grounding, ML, and DL

8.2 Lessons Learned from Real Deployments

The most instructive material came from auditing a reference implementation of a comparable system, read in full before deciding what this project should inherit. Its computer-vision core was sound and worth porting. Its decision layer was largely fabricated, and the fabrications shared one root cause.

A detector that never ran, and a heuristic that filled the gap. The object detector was configured with an invalid model name, so it threw on every load and real detection never executed once. What filled the gap was a luminance heuristic: if the left or right third of the frame deviated more than 60% from mean brightness for three consecutive frames, it logged a phone or book detection — labelled *camera verified*. A window, a desk lamp, or a light-coloured shirt was sufficient. Those fabricated events then fed an escalating penalty system. Compounding it, the load-failure handler set a status string reading *proctoring engine ready*, and that status was never rendered anywhere, so a total detector outage was invisible.

Table 17. Policies converted into structural guarantees.

Policy	Structure
AI must not author claims	Import boundary plus a CI architecture test
Unmeasured is not passed	A result type with no similarity field
No composite score	Vocabulary ban that fails the build
No demographic inference	Classifier module never loaded
No outcome labels	No such dataset and no code path to build one
Plans cannot be back-edited	Question plans are never persisted
Sessions must be re-constructible	Append-only event table with a sequencing trigger

Table 18. Barriers to integration and mitigations.

Barrier	Mitigation
Interoperability	Typed contracts between surfaces; a failed measurement has its own representation rather than a sentinel value
Lifecycle	Export gates refuse to write an artifact that misses its quality bars; artifacts declare their own provenance and the loader rejects one that misrepresents it
Latency budget	On-device scoring for anything on the interactive path; the cheapest model that is not significantly worse
Compliance	Evidence retained per claim; no aggregate to explain because none is produced
Drift	Thresholds and calibration are dated and re-measurable; nothing is authored once and trusted indefinitely

A confident pass manufactured from missing data. The identity endpoint returned a verified result with a similarity of 98.7 and a confidence of 0.98 when no enrolled face profile existed at all. The absence of a record produced a high-confidence pass rather than an error.

Analysis that measured nothing. A résumé analyser substring-matched a fixed keyword list and, on zero matches, returned a plausible-looking hardcoded skill list; its score was floored so an unrelated résumé could never fall below a fifth of the scale. A speech-analytics component took text rather than audio and derived fluency metrics from it.

Table 22 records an argument worth stating explicitly. A proposed live dashboard included confidence, stress, and eye-contact percentages feeding back into question selection. We did not wire them in, and the reasoning generalises: a stress percentage that silently changes which questions a candidate is asked is a hidden score, and one the candidate can neither see nor contest. Every adaptive behaviour that dashboard promised is reachable from what the candidate *said*, which is observable, quotable, and defensible in a rejection conversation.

Table 19. Mapping regulatory obligations onto architectural mechanisms.

Obligation	Instrument	Mechanism in this system
Bias audit of the tool	NYC Local Law 144	No composite score to audit; the audited object is the per-claim verdict distribution
Candidate notification	NYC Local Law 144; Illinois AIVIA	Consent captured per session with a recorded consent version
Human oversight	EU AI Act, Annex III high-risk	Report generation is a deterministic transform containing no code that could emit a recommendation
Explanation and contestability	EU AI Act; HB 3773	Every verdict carries its evidence and character offsets into the candidate’s own words
Non-discrimination	HB 3773; Illinois Human Rights Act	No demographic inference, no outcome labels, no historical decision pattern to inherit
Traceability	All	Append-only event store; sessions reconstructible turn by turn

Table 20. Deployment case studies: where each technology family is exercised and what it produced.

Case	Family	What was done	Core impact
Corpus-scale gate evaluation	Grounding	5,200 synthetic résumés, roughly 100,000 adversarial trials, run offline against real pipeline modules	Surfaced a silent 0.82% false-rejection defect invisible to the unit tests
Threshold calibration	DL	Equal-error sweep on LFW, 2,200 train and 1,000 held-out pairs	False rejection reduced from 0.414 to 0.034
Nine-model comparison	ML	Ten grouped folds, Friedman plus Holm-corrected Wilcoxon, synthetic positive control	Established that algorithm choice is not the bottleneck on this task
Within-query diagnostic	ML	4,112 within-résumé pairs, entity effects cancelled by construction	Separated matching from candidate ranking; refuted our own hypothesis
Sufficiency pipeline validation	ML	Planted-weight synthetic generator with two weightless noise features	Pipeline recovered the planted structure and correctly ignored the noise features
Independent security audit	All	Structured review of 17 public and 23 authenticated endpoints, 3 upload paths	10 findings (5 high, 5 medium); 9 fixed, 1 mitigated

Table 21. Lessons carried into this system’s architecture.

Lesson	Architectural response
A fallback that fabricates is worse than an outage	A failed measurement has its own type carrying no value; nothing can substitute for it
Silent failure is the dangerous failure	Component health is surfaced where a human sees it; corpus-scale evaluation with an explicit true-positive metric
Plausible output invites trust	Outputs are traceable to source spans, so plausibility is checkable rather than persuasive
Hidden weights become verdicts	The composite score is removed, not improved
Reasoned is not correct	Security-adjacent constants require calibration studies with reported error rates

Table 22. Where deep learning is and is not used, and why.

Domain	Decision
Face detection and embedding	Used, pretrained, frozen — genuinely a perception problem
Semantic comparison	Used for clustering and relational features; similarity is not treated as entailment
Claim proposal	Used, but gated; the model never authors
Affect or emotion inference	Not used. No defensible accuracy on a diverse candidate pool, and it maps onto disability, neurodivergence, and cultural difference in expression
Demographic attributes	Never loaded. The bundled classifier is excluded by an allow-list
Deception or cheating estimation	Not built. A learned cheating score has no ground truth and cannot be calibrated

Table 23. Research areas in evidence-based screening: core challenges and key questions.

Research area	Core challenge	Key question
Deployment-aware audit architectures	Continuous provenance without continuous surveillance	How much identity assurance is obtainable from intermittent, consented checks with explicit unmeasured intervals?
Explainable and contestable screening	Explanations that support appeal, not just interpretation	Does a claim audit change human decisions relative to a score, measured by inter-reviewer agreement?
Privacy-preserving process evidence	Keystroke and timing data are highly identifying	Can within-session process deviation be measured on-device so only derived scores leave the client?
Within-query evaluation standards	Pooled metrics conflate capabilities	Should conditioning on the recurring entity become a reporting requirement for paired benchmarks?
Benchmark provenance	Undocumented corpora beget undocumented derivatives	What minimum provenance metadata should a hiring benchmark carry before it is usable for published comparison?
Calibrated abstention	Choosing an operating point for “unknown”	What abstention rate maximises usefulness while preserving honesty?

8.3 Honest Position on Maturity

Three gaps are stated rather than concealed. Identity matching is implemented and unit-tested but is currently only a presence gate on the production candidate path, so the system does *not* today perform continuous identity verification. The face threshold is calibrated on a public benchmark rather than on candidates. No structured recruiter validation interviews have been conducted, which makes the premise that recruiters prefer a claim audit to a score the project’s largest untested assumption. The résumé–posting fit model reached 0.657 AUROC in earlier work and is *not deployed*; no runtime module imports it, and §5 explains why that number is weak.

9 Future Directions and Research Opportunities

9.1 Deployment-Aware Audit Architectures

The immediate engineering task is wiring the calibrated matcher into the live candidate session, which §8 records as not yet done. The research question underneath it is more interesting: continuous verification and candidate privacy pull in

opposite directions, and the honest resolution is probably not more surveillance but better accounting for intervals in which nothing was measured.

9.2 Explainable and Contestable Screening

The project’s core premise — that a claim audit is more useful to a reviewer than a score — is untested. A controlled comparison of reviewers given an audit versus a score, measured by inter-reviewer agreement on identical sessions, would test it directly. We regard this as the single most valuable study the project does not yet have, and note that it could refute the premise.

9.3 Privacy-Preserving Process Evidence

Process signals are informative and dangerous. Our position is that a deviation in how an answer was produced should select a *question*, never raise a *flag* — the candidate’s answer is the evidence. Whether that survives contact with real candidates is untested. Technically, the promising direction is computing deviation on-device so that raw timings never leave the client and only a small number of derived scores are transmitted, which is a privacy property rather than an optimisation.

9.4 Within-Query Evaluation Standards

§5.4 showed that a representation change worth +0.041 pooled AUROC delivered no measurable within-query improvement. The diagnostic costs one pass over predictions already computed, needs no additional labels, and has a fixed null of 0.5. We suggest reporting it alongside pooled metrics as a matter of course for any paired benchmark with a recurring entity, and treating a divergence between the two as information rather than as an anomaly to explain away.

10 Conclusion

CogniHire replaces the composite hiring score with a per-claim audit. Claims are extracted under a grounding constraint enforced by an architectural boundary rather than by prompt discipline, probed in a live interview, and reported as verdicts traceable to evidence — including explicit statements about what was not measured. A human decides.

The methodological contribution is the repeated conversion of policies into structures. “The AI must not author claims” becomes an import boundary with a failing test. “Unmeasured is not passed” becomes a type with no field to hold a fabricated value. “No demographic inference” becomes a module that is never loaded. These are verifiable by inspection in seconds, and they do not decay as attention moves elsewhere.

Three empirical findings stand out. The grounding gate rejected 100% of paraphrases and negation traps across roughly 100,000 adversarial trials while admitting 100% of verbatim claims, making “the model selects, never authors” a measured property rather than a design aspiration. A carefully reasoned similarity constant of 0.50 would have falsely rejected 41.4% of genuine identity matches; measurement reduced that to 3.4%. And nine classifiers spanning six families proved statistically indistinguishable at the top on a public screening benchmark, while a within-query diagnostic showed that the pooled metric’s apparent gains came from ranking candidates rather than matching them to roles.

That last finding deserves the final word, because it began as a different one. The evidence initially indicated that the benchmark’s labels were not relational at all, and a direct test refuted it: four-fifths of the label variance is within-résumé and models recover a real part of it. What survived was narrower — a decoupling between leaderboard progress and task progress that no pooled metric can reveal. We report the sequence rather than the conclusion alone, because a diagnostic capable of distinguishing two hypotheses is only shown to have that power when it actually overturns one.

We report a weak result and a rejected component alongside strong ones. A project that publishes only its successes offers no evidence that it evaluates itself honestly — and honest evaluation is precisely what this system claims to provide to the people it screens.

Data and Code Availability

All analysis code, per-fold results, and figure sources accompany this article, including the experiment scripts, the JSON reports they emit, and the per-(model, fold) CSVs underlying every table and figure. The résumé-posting corpus analysed is publicly available on the Hugging Face Hub; as §5 records, it carries no licence statement, and we analysed it as distributed and redistribute none of it.

No human subjects were involved in the studies reported here. No résumé, job posting, or candidate record was collected, contacted, or re-identified; the face-verification study uses the public LFW benchmark [8], and the grounding evaluation uses a synthetic résumé corpus.

Declaration of Competing Interests

The author declares no competing financial or non-financial interests. This work received no external funding and was not commissioned, sponsored, or reviewed by any vendor of hiring software.

On the Use of AI Assistance

Experiment code, figure generation, and manuscript drafting were carried out with the assistance of a large language model. All experiments were executed locally and their outputs verified against the emitted JSON reports; every citation was retrieved and checked against a primary source before inclusion. The author is responsible for the content, the claims, and any errors.

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